ZisK's Proving System[1]

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Abstract

This document specifies the argument system that ZisK uses for every proof it produces, and analyses its soundness. The protocol is a STARK in the DEEP-ALI + FRI family: a multi-stage algebraic commitment to the execution trace, a random aggregation of the constraints into a single quotient, an out-of-domain consistency check, and a low-degree test of the resulting batch by FRI, compiled into a non-interactive argument by the Fiat–Shamir transform in the random-oracle model.

The specification is parametrised by a configuration record, and the whole of ZisK — the AIR proofs of the execution segments, the leaf verifiers, the aggregation nodes and the Final Verifier — is this one protocol instantiated at different configurations. What distinguishes the layers is the AIR they are given, not the protocol that proves it. The analysis is therefore carried out once, for an arbitrary AIR and configuration, and the recursion is left to instantiate it.

[1]This document describes ZisK v0.16.1 [48cf7cc – 2026-20-03].
Source code available at `https://github.com/0xPolygonHermez/zisk`.

Contents

1. Preliminaries **2**
- 1.1. Notation and Parameters . . . 2
- 1.2. Arithmetization . . . 2
- 1.3. Polynomial Encoding . . . 4
- 1.4. Vector Commitment . . . 5
- 1.5. The Configuration . . . 5

2. The Protocol **6**
- 2.1. What the Protocol Proves . . . 6
- 2.2. Setup . . . 6
- 2.3. The Interactive Protocol . . . 7
- 2.4. FRI . . . 8
- 2.5. Fiat-Shamir . . . 8
- 2.6. The Verifier . . . 9
- 2.7. Commit-and-Prove . . . 9
- 2.8. Deviations . . . 9

3. Soundness **10**
- 3.1. Round-by-Round Soundness . . . 10
- 3.2. The State Function . . . 10
- 3.3. The Round Bounds . . . 11
- 3.4. From Round-by-Round Soundness to Knowledge Soundness . . . 12
- 3.5. Provenance . . . 12

References **14**

1. Preliminaries

This section fixes the notation, recalls the arithmetization that the protocol proves, and adds the two layers the protocol needs on top of it: the encoding of a witness as a Reed–Solomon codeword, and the vector commitment used to commit to such codewords. It closes with the configuration record that parametrises every instance of the protocol.

1.1. Notation and Parameters.

Let $\mathbb{F}$ be a prime field of order p and let $\mathbb{K}$ be an extension of $\mathbb{F}$ of degree d, large enough that $|\mathbb{K}| = p^d$ is exponential in the security parameter λ; all verifier challenges are drawn from $\mathbb{K}$ and all soundness errors below are measured against $|\mathbb{K}|$. For integers $a \le b$ we write $[a, b]$ and $[a, b)$ for the closed and half-open integer intervals and $[b] := [1, b]$. "PPT" means probabilistic polynomial time in λ, and $\mathsf{negl}(\lambda)$ and $\mathsf{poly}(\lambda)$ denote negligible and polynomially bounded functions.

The protocol is analysed in the random-oracle model. We assume a single random oracle $\mathcal{O}$, used for the leaves and the internal nodes of the vector commitment, for the Fiat–Shamir transcript and for the proof-of-work check; the implementation instantiates these roles with separate hash functions, and we treat them as domain-separated instances of $\mathcal{O}$ (Remark 1.1). We write Q for a bound on the total number of oracle queries an adversary makes.

1.2. Arithmetization.

The protocol proves that a witness satisfies an AIR.

Definition 1.1 (AIR and Witness). An *algebraic intermediate representation* (AIR) of width W, depth D and V values is a finite set of *constraint polynomials*

$$\mathcal{C} \subset \mathbb{F}\Big[X_1^{(0)}, ..., X_W^{(0)}, ..., X_1^{(D)}, ..., X_W^{(D)}, Y_1, ..., Y_V\Big].$$

A *witness* of height N for $\mathcal{C}$ is a pair $w = (\boldsymbol{T}, \boldsymbol{a})$ with $\boldsymbol{T} \in \mathbb{F}^{N \times W}$, the *trace*, and $\boldsymbol{a} \in \mathbb{F}^V$, the *AIR values*. At row $r \in [N]$ the variable $X_j^{(k)}$ takes the value $T_j(r + k)$, with row indices taken modulo N, and Y_j takes the value a_j. The witness *satisfies* $\mathcal{C}$ if every constraint polynomial evaluates to zero at every row.

Definition 1.2 (Interaction). An *interaction* of an AIR $\mathcal{C}$ on the bus $b \in \mathbb{F}$ is a tuple $\iota = (b, m_\iota, \boldsymbol{e}_\iota)$, where the *multiplicity* m_ι and the *message* components $e_{\iota,1}, ..., e_{\iota,\ell_b}$ are polynomials over $\mathbb{F}$ in the variables of $\mathcal{C}$. Given a witness w and a row r, we write $m_\iota(w, r) \in \mathbb{F}$ and $\boldsymbol{e}_\iota(w, r) \in \mathbb{F}^{\ell_b}$ for their evaluations at row r. The set of bus identifiers in use is $\mathcal{B}$; every interaction on bus b has the common message length ℓ_b, and $\ell := \max_{b \in \mathcal{B}} \ell_b$.

Throughout, we fix a positive integer M and an indexed family of *AIR types*

$$\mathfrak{A} = \{(\mathcal{C}_\tau, \mathcal{I}_\tau)\}_{\tau \in [M]},$$

where each $\mathcal{C}_\tau$ is an AIR of width W_τ, depth D_τ and V_τ values and $\mathcal{I}_\tau$ is a finite set of interactions of $\mathcal{C}_\tau$. Given a fixed trace height N_τ, we write $\mathcal{W}_\tau := \mathbb{F}^{N_\tau \times W_\tau} \times \mathbb{F}^{V_\tau}$ for the witness space of type τ and fix the public-statement space $\mathcal{P} := \mathbb{F}^k$.

Definition 1.3 (Column Classes and Public Bindings). The description of an AIR $\mathcal{C}$ of width W and V values, used at height N, partitions its column indices as $[W] = J^{\mathrm{fix}} \sqcup J^{\mathrm{cach}} \sqcup J^{\mathrm{wit}}$ and carries:

1. The *fixed columns* J^{fix}, whose values are prescribed as part of the description of $\mathcal{C}$ and hence known to both the prover and the verifier.
2. The *cached columns* J^{cach}, whose values are computed once and reused across different proofs. Their values are not part of the description; instead, a commitment to them (Definition 1.9) travels in the public statement at the coordinates designated by a *cached-commitment binding* $\gamma : [m] \to [k]$.

3. The *witness columns* J^{wit}, computed afresh for every proof and only known in full to the prover.
4. A *public binding map* $\beta : \mathrm{dom}(\beta) \to [k]$ from a set $\mathrm{dom}(\beta) \subset [V]$ to coordinates of the public statement.

The *prescribed data* of $\mathcal{C}$ is the family $\boldsymbol{d} = \big(d_j\big)_{j \in J^{\mathrm{fix}}}$ of fixed columns $d_j \in \mathbb{F}^N$. For a trace $\boldsymbol{T}$ and a set of columns J we write $\boldsymbol{T}|_J$ for the submatrix of the columns in J.

Each AIR type carries this data; we index it by the type, as in J^{fix}_τ, $\boldsymbol{d}_\tau$ and β_τ.

Definition 1.4 (Local Satisfaction). For $\tau \in [M]$, $\boldsymbol{p} \in \mathcal{P}$ and $w = (\boldsymbol{T}, \boldsymbol{a}) \in \mathcal{W}_\tau$, the predicate $\mathrm{Local}_\tau(\boldsymbol{p}, w)$ holds if and only if

1. w satisfies $\mathcal{C}_\tau$.
2. $\boldsymbol{T}$ agrees with the prescribed data on the fixed columns: $T_j(r) = d_{\tau,j}(r)$ for every $j \in J^{\mathrm{fix}}_\tau$ and $r \in [N_\tau]$.
3. $\mathrm{Bind}_\tau(\boldsymbol{p}, w)$: $a_j = p_{\beta_\tau(j)}$ for every $j \in \mathrm{dom}(\beta_\tau)$.
4. The cached columns open the commitment carried in $\boldsymbol{p}$:

$$\mathsf{Commit}_{\mathrm{cach}}\Big(\boldsymbol{T}|_{J^{\mathrm{cach}}_\tau}\Big) = \Big(p_{\gamma_\tau(1)}, \ldots, p_{\gamma_\tau(m)}\Big).$$

Throughout this document we work with a single AIR type and suppress the index τ, writing $\mathcal{C}$, N, W, D, V, $\boldsymbol{d}$, β and $\mathrm{Local}(\boldsymbol{p}, w)$. The recursion instantiates what follows once per type.

Definition 1.1 asks that every constraint polynomial vanish at every row. This is the only *boundary condition* we consider: constraints that are imposed on a strict subset of the rows, such as a first-row or last-row condition, are expressible by multiplying the constraint by a fixed selector column, and the protocol below treats them no differently.

Remark 1.1 (One oracle). *The implementation uses four separately instantiable hash functions: one for the leaves of a Merkle tree, one for its internal nodes, one for the Fiat–Shamir transcript, and a fixed-width one for the proof of work. Modelling them as one oracle is a simplification. It is sound if the four are domain-separated instances of a single function, and it is the modelling under which the bounds below should be read.*

1.2.1. *Stages and challenges.*

The AIR of Definition 1.1 is static: its constraints are polynomials in the trace variables and the AIR values alone. The arguments ZisK builds on top of an AIR — the lookup and permutation arguments that implement the buses [Hab22] — need columns that the prover can only compute *after* seeing a verifier challenge, and constraints that mention that challenge. The protocol therefore proves a *staged* AIR.

Definition 1.5 (Staged AIR). A *staging* of an AIR $\mathcal{C}$ into $S \geq 1$ stages consists of a partition of the witness columns

$$J^{\mathrm{wit}} = J^{(1)} \sqcup \cdots \sqcup J^{(S)}$$

together with *challenge counts* $c_1, \ldots, c_{S-1} \in \mathbb{N}$. Writing $c := c_1 + \cdots + c_{S-1}$, the constraint polynomials of a staged AIR are allowed to depend on c further variables $Z_1, \ldots, Z_c$, the *challenge variables*, which take values in $\mathbb{K}$; consequently the constraints and the AIR values of a staged AIR take values in $\mathbb{K}$ rather than in $\mathbb{F}$. The columns of the first stage are the ones that hold the execution trace proper; they depend on no challenge.

A *challenge vector* is a tuple $\chi = \big(\chi^{(1)}, \ldots, \chi^{(S-1)}\big)$ with $\chi^{(s)} \in \mathbb{K}^{c_s}$. For a fixed χ, binding the challenge variables to χ turns the staged AIR into an ordinary AIR over $\mathbb{K}$, and we write $\mathrm{Local}(\boldsymbol{p}, w; \chi)$ for the predicate of Definition 1.4 of that AIR. The staging constrains the prover, not the predicate: the columns of $J^{(s)}$ must be determined before $\chi^{(s)}$ is drawn, and may depend on $\chi^{(1)}, \ldots, \chi^{(s-1)}$.

In ZisK, $S = 2$: the first stage holds the execution trace, the second holds the auxiliary columns of the bus arguments, and $c_1 = 2$, the two challenges being the LogUp pair. We keep S and the c_s general, since the compiler emits them per circuit.

1.3. **Polynomial Encoding.**

Fix a trace height $N = 2^n$ and assume 2^n divides $p - 1$, so that $\mathbb{F}^*$ has a subgroup

$$\Omega = \langle \omega \rangle \subset \mathbb{F}^*, \quad |\Omega| = N,$$

of order N with generator ω. We identify the row index $r \in [N]$ with the point $\omega^r \in \Omega$; the row shift $r \mapsto r + 1$ of Definition 1.1 becomes multiplication by ω, and row indices taken modulo N become the group structure of Ω.

Definition 1.6 (Trace polynomials and the constraint polynomial). Let $w = (\boldsymbol{T}, \boldsymbol{a})$ be a witness of height N. For every column $j \in [W]$ let $\hat{T}_j \in \mathbb{K}[X]$ be the unique polynomial of degree less than N with $\hat{T}_j(\omega^r) = T_j(r)$ for all $r \in [N]$. For a constraint polynomial $C \in \mathcal{C}$ and a challenge vector χ, the *constraint polynomial* of C on w is

$$\hat{C}(X) \coloneqq C\Big(\hat{T}_1(X), ..., \hat{T}_W(X), ..., \hat{T}_1(\omega^D X), ..., \hat{T}_W(\omega^D X), \boldsymbol{a}, \chi\Big) \in \mathbb{K}[X],$$

obtained by substituting $\hat{T}_j\big(\omega^k X\big)$ for the variable $X_j^{(k)}$, the AIR values for the variables Y, and χ for the challenge variables Z.

Write $Z_\Omega(X) \coloneqq X^N - 1$ for the vanishing polynomial of Ω. Since $\hat{C}(\omega^r)$ is exactly the evaluation of C at row r, the satisfaction condition of Definition 1.1 becomes a divisibility condition.

Definition 1.7 (Quotient). Let χ be a challenge vector and $\zeta \in \mathbb{K}$. The *aggregated constraint* of $\mathcal{C} = \Big\{C_1, ..., C_{|\mathcal{C}|}\Big\}$ at ζ is

$$\hat{C}_\zeta(X) \coloneqq \sum_{i=1}^{|\mathcal{C}|} \zeta^{i-1} \cdot \hat{C}_i(X),$$

and, when Z_Ω divides $\hat{C}_\zeta$, the *quotient* is $Q(X) \coloneqq \hat{C}_\zeta(X)/Z_\Omega(X)$.

Remark 1.2 (Satisfaction as divisibility). *A witness w satisfies $\mathcal{C}$ under χ if and only if Z_Ω divides $\hat{C}_i$ for every i, and then Z_Ω divides $\hat{C}_\zeta$ for every ζ. Conversely, if some $\hat{C}_i$ is not divisible by Z_Ω, then Z_Ω divides $\hat{C}_\zeta$ for at most $|\mathcal{C}| - 1$ values of ζ, since divisibility is a linear condition and the map $\zeta \mapsto \big(1, \zeta, ..., \zeta^{|\mathcal{C}|-1}\big)$ meets any proper subspace in at most $|\mathcal{C}| - 1$ points. Aggregating with the* powers *of one challenge rather than with $|\mathcal{C}|$ independent challenges is what makes this count linear in $|\mathcal{C}|$ rather than a single inverse power of $|\mathbb{K}|$; the implementation aggregates with powers.*

If every constraint has total degree at most δ in the trace variables, then $\deg \hat{C}_\zeta < \delta N$ and $\deg Q < (\delta - 1)N$. The quotient is committed in $q \coloneqq \delta - 1$ chunks of degree less than N,

$$Q(X) = \sum_{i=0}^{q-1} X^{iN} \cdot Q_i(X), \quad \deg Q_i < N,$$

so that all committed polynomials have the same degree bound.

1.3.1. *The evaluation domain.*

Definition 1.8 (Reed-Solomon code). Let $\mathcal{D} \subset \mathbb{K}$ be a set with $|\mathcal{D}| > N$. The *Reed–Solomon code* $\mathrm{RS}[\mathcal{D}, N]$ is the set of functions $f : \mathcal{D} \to \mathbb{K}$ that agree with a polynomial of degree less than N. Its *rate* is $\rho \coloneqq N/|\mathcal{D}|$, and the relative Hamming distance of f from the code is written $\Delta(f, \mathrm{RS}[\mathcal{D}, N])$.

The protocol evaluates the trace polynomials on a domain disjoint from Ω. Fix a blowup exponent $\nu \geq 1$, put $n_{\text{ext}} := n + \nu$, assume $2^{n_{\text{ext}}}$ divides $p - 1$, let ω_{ext} generate the subgroup of $\mathbb{F}^*$ of order $2^{n_{\text{ext}}}$, and fix a shift $\eta \in \mathbb{F}^*$ outside that subgroup. The *evaluation domain* is the coset

$$\mathcal{D} := \eta \cdot \langle \omega_{\text{ext}} \rangle, \quad |\mathcal{D}| = 2^{n_{\text{ext}}} = N/\rho, \quad \rho = 2^{-\nu},$$

which is disjoint from Ω, so the quotient of Definition 1.7 may be evaluated on all of $\mathcal{D}$ without dividing by zero. The *low-degree extension* of a column T_j is the restriction of $\hat{T}_j$ to $\mathcal{D}$; it is a codeword of $\mathrm{RS}[\mathcal{D}, N]$, and the prover commits to it rather than to the column itself.

1.4. Vector Commitment.

Definition 1.9 (Deterministic Commitment Scheme). A *deterministic commitment scheme* Com consists of a set of *messages*, a finite set of *commitments*, and a polynomial-time computable function Commit from the former to the latter. It is *binding* if for every PPT adversary $\mathcal{A}$

$$\mathsf{Adv}^{\mathsf{bind}}_{\mathsf{Com}}(\mathcal{A}) := \Pr\big[(m, m') \leftarrow \mathcal{A}\big(1^\lambda\big) : m \neq m' \wedge \mathsf{Commit}(m) = \mathsf{Commit}(m')\big] \leq \mathsf{negl}(\lambda).$$

Since Commit is deterministic, binding is collision resistance of Commit. When the messages are vectors or matrices over $\mathbb{F}$, the scheme is a *vector commitment* if it also offers openings of individual positions.

The protocol uses one such scheme, a Merkle tree over the random oracle, applied to matrices whose rows are indexed by the evaluation domain.

Definition 1.10 (Merkle commitment). Fix an *arity* $a \geq 2$ and a *cap level* $\ell \geq 0$. The Merkle commitment to a matrix $\boldsymbol{M} \in \mathbb{K}^{|\mathcal{D}| \times u}$ hashes each row to a leaf, $\mathcal{O}\big(M_{x,1}, \ldots, M_{x,u}\big)$ for $x \in \mathcal{D}$, and then repeatedly hashes groups of a consecutive nodes until a level of at most a^ℓ nodes is reached; the commitment is the hash of that level. An *opening* at $x \in \mathcal{D}$ consists of the row $\boldsymbol{M}_x$, the $a - 1$ siblings at each level above the leaf and below the cap, and the whole capped level.

Taking $\ell = 0$ gives the usual Merkle tree; a positive cap level shortens the authentication path at the cost of transmitting the capped level once per proof, and the verifier recomputes the commitment from it. Committing to a matrix rather than to a vector means that all columns of one stage are opened together at a queried point, which is what the protocol needs.

We will use that a Merkle commitment over a random oracle is *straight-line extractable*: from the list of oracle queries made before a commitment is announced, one can reconstruct the only matrix the announcer can later open it to, except with probability $O(Q^2/|\mathbb{K}|)$.

1.5. The Configuration.

Every instance of the protocol is determined by an AIR together with the following record. It is exactly the data the setup emits per circuit.

Symbol	Meaning	Implementation
n	trace height exponent, $N = 2^n$	`n_bits`
ν	blowup exponent, $\rho = 2^{-\nu}$	`n_bits_ext` − `n_bits`
S, $(c_s)_s$	stages and challenges per stage (Definition 1.5)	`n_stages`, `challengesMap`
δ	maximum constraint degree; $q = \delta - 1$ quotient chunks	`air_max_degree`, `q_deg`
Γ	opening offsets, a finite subset of $\mathbb{Z}$	`opening_points`
a, ℓ	Merkle arity and cap level (Definition 1.10)	`arity`, `last_level_verification`
L, $(m_0, \ldots, m_L)$	number of FRI layers and their size exponents	`n_fri_steps`, `fri_steps`
t	number of FRI queries	`n_fri_queries`

Symbol	Meaning	Implementation
b	proof-of-work bits	`pow_bits`
θ	proximity parameter of the low-degree test	`proximity_gap`

TABLE 1. The configuration record cfg.

As an illustration of the orders of magnitude, ZisK's Final circuit is configured with $n = 16$ and $\nu = 4$, so $\rho = 1/16$; $S = 2$ stages over 46 fixed columns; $\delta = 8$, hence $q = 7$ quotient chunks; $\Gamma = \{-1, 0, 1, 2\}$; Merkle arity $a = 4$ capped at $\ell = 2$; FRI layer exponents $(20, 16, 12, 8, 5)$, so $L = 4$ layers folded with arities $16, 16, 16, 8$; $t = 54$ queries; and $b = 22$ grinding bits. The other circuits differ in every one of these numbers.

The record contains one further Boolean, recorded here for completeness: the transcript either absorbs the public inputs, the out-of-domain evaluations and the final FRI polynomial directly, or absorbs a hash of each. The choice changes the Fiat–Shamir transcript but none of the checks, and we fix the first alternative in the description below.

2. The Protocol

We now specify the argument system. It is presented in two steps: first as an interactive protocol between a prover and a verifier, in which the verifier's messages are uniform elements of $\mathbb{K}$ and the prover answers queries to committed vectors, and then as the non-interactive argument obtained by deriving the verifier's messages from the transcript with the random oracle.

Throughout, fix a staged AIR $\mathcal{C}$ with S stages (Definition 1.5) and a configuration cfg (Table 1). Recall from Section 1.3 that $N = 2^n$ is the trace height, $\Omega = \langle\omega\rangle$ the trace domain, $\mathcal{D} = \eta\langle\omega_{\mathrm{ext}}\rangle$ the evaluation domain of size $2^{n_{\mathrm{ext}}} = N/\rho$, that $Z_\Omega(X) = X^N - 1$, and that the quotient is split into $q = \delta - 1$ chunks.

2.1. What the Protocol Proves.

Definition 2.1 (AIR satisfaction relation). For a staged AIR $\mathcal{C}$ with prescribed data $\boldsymbol{d}$ and bindings β, γ, the *AIR satisfaction relation* is

$$\mathcal{R}_{\mathcal{C}} := \{((\boldsymbol{p}, \chi); w) \mid w \in \mathcal{W} \wedge \mathrm{Local}(\boldsymbol{p}, w; \chi)\},$$

where $\boldsymbol{p} \in \mathcal{P}$ is a public statement, χ a challenge vector and $w = (\boldsymbol{T}, \boldsymbol{a})$ a witness of height N.

Two features of this relation deserve comment, because they are what make it usable by a caller that composes several proofs.

First, the challenge vector χ is part of the *statement*, not of the witness, and not something the protocol samples. The relation is a family indexed by χ, and an argument system for it says nothing about where χ came from. Section 2.5 describes the two ways the implementation supplies it, and Remark 2.1 explains why the distinction is not cosmetic.

Second, a caller may need the proof to be bound to the prover's commitment to the trace, rather than only to the public statement. The protocol commits to the stage-1 columns before anything else happens, so its first message is such a commitment; a caller that needs it can read it off the proof and treat it as an additional public output. We write rt_1 for it below and return to this in Section 2.7.

2.2. Setup.

The setup is deterministic and depends only on $\mathcal{C}$ and cfg. It computes the low-degree extensions of the fixed columns, that is the restrictions to $\mathcal{D}$ of the interpolants of $\boldsymbol{d}$, commits to them with Definition 1.10, and outputs

$$\mathsf{vk} := (\mathrm{rt}_{\mathcal{C}}, \mathcal{C}, \mathrm{cfg}),$$

where $\mathrm{rt}_{\mathcal{C}}$ is that commitment. The verification key contains no secret and no randomness: it is a digest of the circuit description, and two parties that agree on $\mathcal{C}$ and cfg compute the same vk. Cached columns are *not* part of it; they are bound through the public statement (Definition 1.4).

2.3. The Interactive Protocol.

The prover holds $(\boldsymbol{p}, \chi)$ and a witness w with $\mathrm{Local}(\boldsymbol{p}, w; \chi)$. All prover messages are either Merkle commitments to matrices over $\mathcal{D}$, elements of $\mathbb{K}$, or openings; all verifier messages are uniform elements of $\mathbb{K}$.

1. **Stage commitments.** For $s = 1, \ldots, S$: the prover computes the columns of stage s, extends them to $\mathcal{D}$, and sends the Merkle commitment rt_s to the matrix whose rows are the values of those columns at the points of $\mathcal{D}$, together with the AIR values of stage s. For $s < S$ the verifier replies with $\chi^{(s)} \in \mathbb{K}^{c_s}$, which the stage $s+1$ columns may use. After the last stage the verifier replies with the *aggregation challenge* $\zeta \in \mathbb{K}$.
2. **Quotient.** The prover forms the aggregated constraint $\hat{C}_\zeta$ of Definition 1.7, divides by Z_Ω, splits the quotient into the q chunks $Q_0, \ldots, Q_{q-1}$ of degree less than N, extends them to $\mathcal{D}$ and sends the Merkle commitment rt_{S+1} to them. The verifier replies with an *out-of-domain point* $\xi \in \mathbb{K}$.
3. **Out-of-domain evaluations.** The prover sends the values
$$\hat{f}(\xi\omega^o) \quad \text{for every committed column } f \text{ and every } o \in \Gamma,$$
and the values $\hat{Q}_0(\xi), \ldots, \hat{Q}_{q-1}(\xi)$. Here $\Gamma \subset \mathbb{Z}$ is the set of opening offsets of cfg, and a negative offset means $\omega^{-|o|}$. Write $\boldsymbol{v}$ for the tuple of all these values. The verifier replies with *batching challenges* $\boldsymbol{\mu} \in \mathbb{K}^2$.
4. **Low-degree test.** Write $\Gamma = \{o_1, \ldots, o_{|\Gamma|}\}$ for the opening offsets in the order fixed by the circuit description, and, for each i, let $f_{i,1}, \ldots, f_{i,n_i}$ be the committed columns that are opened at $\xi\omega^{o_i}$ — among the fixed columns, the columns of every stage and the quotient chunks. Let $g_0 : \mathcal{D} \to \mathbb{K}$ be the function
$$g_0(x) := \sum_{i=1}^{|\Gamma|} \mu_1^{|\Gamma|-i} \cdot \frac{1}{x - \xi\omega^{o_i}} \cdot \sum_{j=1}^{n_i} \mu_2^{n_i-j} \cdot \big(f_{i,j}(x) - \hat{f}_{i,j}(\xi\omega^{o_i})\big).$$
The prover and the verifier then run the FRI protocol of Section 2.4 on g_0.

Both sums are Horner evaluations in a single challenge: μ_2 batches the columns opened at one point, and μ_1 batches across the opening points. Neither is a random linear combination with independent coefficients, and the nesting means the coefficient of a given column is a product $\mu_1^a \mu_2^b$ rather than a single power; Section 2.8 returns to this. Note also that the columns opened at different offsets need not be the same: an offset is present in Γ because some constraint refers to that row, and only the columns that constraint reads are opened there.

The value $g_0(x)$ is a *DEEP quotient* [Ben+19]: it is a low-degree function precisely when every committed column agrees with a polynomial of degree less than N *and* the claimed out-of-domain values are the evaluations of those polynomials. Sampling ξ from the whole of $\mathbb{K}$, rather than from $\mathcal{D}$, is what lets a single evaluation pin down the committed polynomial; it is the difference between DEEP-FRI and plain FRI, and it is what makes the soundness error depend on $|\mathbb{K}|$ rather than on $|\mathcal{D}|$.

The verifier's algebraic check is separate from the low-degree test, and uses only the out-of-domain values:

Definition 2.2 (Out-of-domain constraint check). Given $\boldsymbol{v}$, the public statement $\boldsymbol{p}$ and the challenges χ, ζ, the verifier evaluates the aggregated constraint $\hat{C}_\zeta$ at ξ by substituting the claimed values $\boldsymbol{v}$ for the trace variables, and accepts this check if

$$\hat{C}_\zeta(\xi) = Z_\Omega(\xi) \cdot \sum_{i=0}^{q-1} \xi^{iN} \cdot \hat{Q}_i(\xi).$$

This is the *algebraic linking identity* of the protocol: it is the only place where the constraints of $\mathcal{C}$ are used, and it is checked at a single point of $\mathbb{K}$. Everything else in the protocol exists to make the values $\boldsymbol{v}$ trustworthy.

2.4. FRI.

FRI [Ben+18] tests that a function $g_0 : \mathcal{D} \to \mathbb{K}$ is close to $\mathrm{RS}[\mathcal{D}, \rho|\mathcal{D}|]$. The configuration fixes layer exponents $m_0 = n_{\mathrm{ext}} > m_1 > \cdots > m_L$; layer i lives on the domain $\mathcal{D}_i$ of size 2^{m_i} obtained by raising $\mathcal{D}$ to the power $2^{m_0 - m_i}$, and folding from layer i to layer $i+1$ has arity $2^{m_i - m_{i+1}}$.

1. **Commit phase.** The prover sends a Merkle commitment to g_0, grouping the $2^{m_0 - m_1}$ values that share a layer-1 preimage into one row, so that a single opening reveals a whole folding group. For $i = 1, \ldots, L$ the verifier sends a folding challenge $\beta_i \in \mathbb{K}$ and the prover sends a commitment to
$$g_i := \mathrm{Fold}_{\beta_i}(g_{i-1}),$$
the unique function on $\mathcal{D}_i$ whose value at y is the evaluation at β_i of the interpolant of the $2^{m_{i-1} - m_i}$ values of g_{i-1} above y. For the last layer the prover sends g_L in the clear.
2. **Query phase.** The verifier picks t indices uniformly in $[0, 2^{m_0})$. For each index it opens the corresponding folding group of every layer, checks each Merkle opening, and checks that folding the group of layer i with β_{i+1} gives the opened value of layer $i+1$, down to g_L.
3. **Final check.** The verifier interpolates g_L and checks that its coefficients of index at least $\rho \cdot 2^{m_L}$ vanish, that is that g_L is a codeword of $\mathrm{RS}[\mathcal{D}_L, \rho 2^{m_L}]$.

Layer 0 is committed even though the verifier can compute g_0 at any queried point from the openings of Section 2.3. The consistency check between the two — the value recomputed from the stage openings against the value opened from the layer-0 commitment — is what ties the low-degree test to the committed trace, and it is performed at every queried index.

2.5. Fiat-Shamir.

The interactive protocol is made non-interactive in the usual way: the verifier's messages are replaced by outputs of $\mathcal{O}$ on the transcript so far. The transcript is seeded, then absorbs each prover message and emits each verifier message in the order of Section 2.3 and Section 2.4, and the query indices are drawn last.

Two features of the compilation are specific to the implementation.

Grinding. After the last folding challenge the transcript emits one further element $c^* \in \mathbb{K}$. The prover must find a nonce $\mathrm{nc} \in \mathbb{F}$ with
$$\mathcal{O}(c^*, \mathrm{nc})_1 < p/2^b,$$
and sends it; the verifier recomputes the check. The t query indices are then derived from (c^*, nc). Grinding raises by b bits the cost of resampling the query set, which is the step an adversary would otherwise repeat cheaply after committing to everything else.

Transcript seeding. There are two modes. In the *standalone* mode the transcript is seeded with vk, the public statement $\boldsymbol{p}$ and the stage-1 commitment rt_1, so that $\chi^{(1)}$ — in ZisK, the LogUp pair — is a function of the committed trace. In the *global* mode the transcript is seeded with an externally supplied $\chi^{(1)}$ and with nothing else; the stage-1 commitment is absorbed afterwards, with the rest of the transcript.

Remark 2.1 (The global mode does not bind the trace to the challenge). *In the standalone mode, $\chi^{(1)}$ is drawn after rt_1 has entered the transcript, so a prover must fix the trace before learning the challenge, and the usual Fiat–Shamir argument applies. In the global mode it is not: $\chi^{(1)}$ seeds the transcript and rt_1 enters it afterwards, so nothing in* this *protocol prevents a prover from choosing the trace as a function of $\chi^{(1)}$.*

This is deliberate and it is what makes proof aggregation possible: every instance in a run must compute its bus contribution under the same *challenge, which cannot happen if each derives its own from its own commitment. The binding that the protocol gives up is supplied from outside, by whatever fixes $\chi^{(1)}$ — in ZisK, by a global challenge derived from an accumulation of all the stage-1 commitments of the run.*

For this document the consequence is a scoping one, and it is the reason Definition 2.1 puts χ in the statement. In the global mode the results below are statements for a fixed χ*, and they carry no claim that χ was chosen independently of the witness. Discharging that is the caller's obligation, and it is discharged where the global challenge is derived, not here.*

2.6. **The Verifier.**

Collecting the checks, the verifier $V(\mathsf{vk}, (\boldsymbol{p}, \chi), \pi) \in \{0, 1\}$ accepts if and only if all of the following hold.

1. The proof parses at the sizes dictated by cfg, and every field element in it is canonical.
2. Replaying the transcript of Section 2.5 on the prover messages in π reproduces the challenges $\chi^{(2)}, ..., \zeta, \xi, \boldsymbol{\mu}, \beta_1, ..., \beta_L$ and c^* used below, and in the standalone mode also $\chi^{(1)}$.
3. The grinding check of Section 2.5 holds for the nonce in π.
4. The out-of-domain constraint check of Definition 2.2 holds.
5. For each of the t query indices: the Merkle openings of the fixed columns, of each stage and of the quotient chunks verify against $\mathrm{rt}_{\mathcal{C}}$, $\mathrm{rt}_1, ..., \mathrm{rt}_S$ and rt_{S+1}; the value of g_0 recomputed from those openings agrees with the value opened from the layer-0 commitment; and the FRI folding checks of Section 2.4 hold down to g_L.
6. The final polynomial g_L has degree less than $\rho 2^{m_L}$.

The verifier reads the witness only through the openings at t points and the $|\boldsymbol{v}|$ out-of-domain values, so its running time is polynomial in cfg and in the size of the AIR description, and independent of N except through the logarithmic depth of the Merkle openings.

2.7. **Commit-and-Prove.**

rt_1 is the first prover message, and it is a Merkle commitment to the stage-1 columns, that is to the execution trace. A caller may therefore treat it as a public output of the proof and impose its own conditions on it. This is how the recursion binds a proof to a trace without re-proving anything: it exposes rt_1 and hashes it, rather than hashing the witness.

Two cautions belong here rather than in the caller. First, rt_1 commits to the stage-1 columns only. Cached columns are committed separately and bound through $\boldsymbol{p}$ (Definition 1.4), and fixed columns through vk; a caller that needs rt_1 to determine the whole witness must check that its AIR places the columns it cares about in stage 1. Second, the commitment is to the *low-degree extension* of the trace, not to the trace itself; the two determine each other, but only for a codeword, which is what the low-degree test establishes and only up to the proximity parameter.

2.8. **Deviations.**

The protocol above is a DEEP-ALI STARK with FRI as its low-degree test, and each of its steps is standard. The following choices are not the textbook ones, and each of them will cost a term or a hypothesis in the analysis; we collect them here so that the analysis can refer to them.

1. **Power batching, at two levels.** The aggregation of the constraints (Definition 1.7) uses powers of one challenge, and the batching of the DEEP quotients uses powers of two, nested: μ_2 within an opening point and μ_1 across opening points (Section 2.3). None of the three is a random linear combination with independent coefficients. For the constraint aggregation the loss is linear in the number of constraints rather than a single inverse power of $|\mathbb{K}|$ (Remark 1.2). For the DEEP batch the relevant proximity-gap statement is the one for low-degree parametrised curves rather than

for affine spaces [Ben+20], and the nesting means the curve has degree $|\Gamma| \cdot \max_i n_i$ in the pair (μ_1, μ_2) rather than degree one in a single challenge.
2. **Non-uniform folding.** The FRI layer exponents are an arbitrary decreasing sequence, so folding arities vary from layer to layer; the usual analyses fix one arity.
3. **Early stopping.** FRI stops at layer L and sends g_L in the clear rather than folding to a constant.
4. **Capped Merkle trees.** Trees have arity a and are capped at level ℓ (Definition 1.10), the capped level being transmitted and the root recomputed from it. Straight-line extraction has to be restated for this variant.
5. **Multiple opening offsets.** Γ is an arbitrary finite set of signed row offsets, so a constraint may refer to rows both above and below the current one.
6. **Externally supplied challenges.** The global mode of Remark 2.1.
7. **One oracle.** The four hash roles are modelled as one (Remark 1.1).

3. Soundness

This section states what is to be proved. The protocol of Section 2 is analysed in two steps, in the order in which it was presented: first the interactive protocol, for which we exhibit a *round-by-round* state function and bound, round by round, the probability that a verifier message rescues a transcript that is already lost; then the Fiat-Shamir compilation, for which the round-by-round bound and the random-oracle model together give knowledge soundness with a straight-line extractor.

We give the definitions and the statements here, together with the provenance of each bound: which are instances of a published theorem, which are instances of one that has to be restated for the variant ZisK uses, and which have no counterpart in the literature. Proofs are deferred.

3.1. Round-by-Round Soundness.

Round-by-round soundness [Can+18] is the property that makes the Fiat-Shamir transform provably sound. It asks for a way of labelling every partial transcript as *live* or *doomed*, such that a false statement starts doomed, a doomed transcript stays doomed under any prover message, a doomed full transcript is rejected, and the only way out is a lucky verifier message. The soundness of the whole protocol is then controlled by a single quantity: how lucky.

Definition 3.1 (State function and round-by-round soundness). Let Π be a public-coin interactive protocol for a relation $\mathcal{R}$, in which the prover's messages may be oracles. A *state function* for Π is a function St assigning to every statement x and every partial transcript tr a value in $\{\mathrm{live}, \mathrm{doomed}\}$, such that:

1. if x has no witness, then $\mathrm{St}(x, \varnothing) = \mathrm{doomed}$;
2. if $\mathrm{St}(x, \mathrm{tr}) = \mathrm{doomed}$ and tr' extends tr by a prover message, then $\mathrm{St}(x, \mathrm{tr}') = \mathrm{doomed}$;
3. if tr is a complete transcript with $\mathrm{St}(x, \mathrm{tr}) = \mathrm{doomed}$, then the verifier rejects.

The *round-by-round soundness error* of Π with respect to St is

$$\varepsilon_{\mathrm{rbr}} := \max_{x,\ \mathrm{tr}} \Pr_c[\mathrm{St}(x, \mathrm{tr} \parallel c) = \mathrm{live}],$$

the maximum over statements x, over partial transcripts tr that are doomed and await a verifier message, of the probability over that message c that the extended transcript becomes live.

The state function need not be efficiently computable; it is a proof device, not an algorithm. For an oracle protocol it may inspect the whole of every oracle the prover has sent, which is what lets it speak about the committed functions rather than about the few positions the verifier opens.

3.2. The State Function.

Fix a statement $(\boldsymbol{p}, \chi)$ and a *proximity parameter* $\theta \in (0, 1 - \sqrt{\rho})$, below the Johnson bound, so that every function within relative distance θ of $\mathrm{RS}[\mathcal{D}, N]$ has a unique closest codeword; for such an f we write $\mathrm{dec}(f)$ for the polynomial of degree less than N that it decodes to. The state function of the protocol is the following; each clause refers to the round of Section 2.3 or Section 2.4 that produces it, and a transcript is doomed as soon as any clause that applies to it holds.

Definition 3.2 (State function of the protocol). The transcript is *doomed* if any of the following holds.

1. **Start.** There is no w with $\mathrm{Local}(\boldsymbol{p}, w; \chi)$.
2. **Stages.** Some committed function — a fixed column, a column of a stage committed so far, or, later, a quotient chunk — is at relative distance at least θ from $\mathrm{RS}[\mathcal{D}, N]$.
3. **Stages.** All committed columns are within θ, and the witness $\mathrm{dec}(w)$ they decode to does not satisfy $\mathcal{C}$ under χ.
4. **Aggregation.** The aggregation challenge ζ has been drawn and Z_Ω does not divide $\hat{C}_\zeta$, the aggregated constraint of $\mathrm{dec}(w)$.
5. **Quotient.** The quotient chunks have been committed and $\hat{C}_\zeta \neq Z_\Omega \cdot \sum_i X^{iN}\, \mathrm{dec}(Q_i)$ as polynomials.
6. **Out of domain.** The point ξ and the values $\boldsymbol{v}$ have been sent, and either the check of Definition 2.2 fails, or some entry of $\boldsymbol{v}$ differs from the evaluation of the corresponding decoded polynomial at its opening point.
7. **Batching.** The batching challenges $\boldsymbol{\mu}$ have been drawn and the function g_0 they determine is at relative distance at least θ from $\mathrm{RS}[\mathcal{D}, N]$.
8. **Folding.** A folding challenge β_i has been drawn and g_i is at relative distance at least θ from $\mathrm{RS}[\mathcal{D}_i, \rho 2^{m_i}]$.

Remark 3.1 (Reading the state function). *Clauses 3 to 6 are the chain that carries a false statement from the algebra to the low-degree test, and each link is a separate lemma below: a witness that does not satisfy the constraints makes the aggregated constraint indivisible (clause 4), an indivisible aggregated constraint admits no quotient (clause 5), and no quotient makes the out-of-domain identity fail at a random point (clause 6). A prover that wants to survive must therefore lie in $\boldsymbol{v}$, and a lie in $\boldsymbol{v}$ is exactly what makes the DEEP quotients of Section 2.3 fail to be low-degree, which clause 7 records and the query phase detects.*

3.3. The Round Bounds.

Each of the following statements bounds the probability that one verifier message turns a doomed transcript live. Together they give Theorem 3.8. We write $|\mathcal{C}|$ for the number of constraints, δ for their maximum degree and $\Lambda := \sum_i n_i$ for the total number of out-of-domain values.

Lemma 3.3 (Aggregation). *Let the committed columns decode to a witness that does not satisfy $\mathcal{C}$ under χ. Then*

$$\Pr_{\zeta \leftarrow \mathbb{K}}\left[Z_\Omega \text{ divides } \hat{C}_\zeta\right] \leq \frac{|\mathcal{C}| - 1}{|\mathbb{K}|}.$$

Lemma 3.4 (Out-of-domain). *Let Z_Ω not divide $\hat{C}_\zeta$, and let the quotient chunks decode to polynomials $\mathrm{dec}(Q_i)$. Then, over a uniform $\xi \in \mathbb{K}$, the probability that the check of Definition 2.2 holds for the* true *evaluations of the decoded polynomials is at most $\delta N / |\mathbb{K}|$.*

Lemma 3.5 (Batching). *Suppose some entry of $\boldsymbol{v}$ differs from the evaluation of the corresponding decoded polynomial at its opening point. Then, over uniform $\boldsymbol{\mu} = (\mu_1, \mu_2) \in \mathbb{K}^2$,*

$$\Pr[\Delta(g_0, \mathrm{RS}[\mathcal{D}, N]) < \theta] \leq \varepsilon_{\mathrm{batch}}(\theta, |\Gamma|, \Lambda).$$

Lemma 3.6 (Folding). *Suppose* $\Delta(g_{i-1}, \mathrm{RS}[\mathcal{D}_{i-1}, \rho 2^{m_{i-1}}]) \geq \theta$. *Then, over a uniform folding challenge* $\beta_i \in \mathbb{K}$,

$$\Pr[\Delta(g_i, \mathrm{RS}[\mathcal{D}_i, \rho 2^{m_i}]) < \theta] \leq \varepsilon_{\mathrm{fold}}(\theta, m_{i-1} - m_i).$$

Lemma 3.7 (Query phase). *Suppose the transcript is doomed when the query indices are drawn. Then the verifier accepts with probability at most* $(1-\theta)^t$ *over the choice of the* t *indices.*

Theorem 3.8 (Round-by-round soundness). *Definition 3.2 is a state function for the interactive protocol of Section 2, and its round-by-round soundness error is*

$$\varepsilon_{\mathrm{rbr}} \leq \max\left\{ \frac{|\mathcal{C}|-1}{|\mathbb{K}|}, \quad \frac{\delta N}{|\mathbb{K}|}, \quad \varepsilon_{\mathrm{batch}}, \quad \varepsilon_{\mathrm{fold}}, \quad (1-\theta)^t \right\}.$$

3.4. From Round-by-Round Soundness to Knowledge Soundness.

The interactive protocol is an oracle protocol, and the compilation of Section 2.5 replaces its oracles by Merkle commitments and its verifier messages by random-oracle outputs. This is the BCS transform [BCS16], whose guarantee is what the recursion consumes.

Proposition 3.9 (Compilation). *Let the interactive protocol have* r *rounds and round-by-round soundness error* $\varepsilon_{\mathrm{rbr}}$. *Then the non-interactive argument of Section 2.5 is knowledge-sound for* $\mathcal{R}_{\mathcal{C}}$ *against adversaries making at most* Q *random-oracle queries, with error*

$$\varepsilon_{\mathrm{ks}} \leq c(Q, r) \cdot \varepsilon_{\mathrm{rbr}} \cdot 2^{-b} + \varepsilon_{\mathrm{mt}}(Q),$$

where $c(Q, r)$ *is the loss of the passage from round-by-round to state-restoration soundness,* 2^{-b} *is the gain from grinding, and* $\varepsilon_{\mathrm{mt}}(Q)$ *collects the Merkle-tree terms. The extractor is straight-line: it reads the committed functions off the adversary's oracle queries, decodes them, and outputs the witness.*

This is the statement that the analysis of the recursion assumes of each of its four argument systems, and the extractor it produces is the same Merkle-tree extractor that that analysis uses for the trace commitment, so its two extractability assumptions are one mechanism rather than two.

Remark 3.2 (What the compilation does not give). *Proposition 3.9 is a statement for a* fixed χ, *because* $\mathcal{R}_{\mathcal{C}}$ *is (Definition 2.1). In the global mode of Remark 2.1 the prover sees* χ *before committing, so the guarantee is only as good as the caller's argument that* χ *was fixed independently of the witness. Nothing in this section supplies that argument, and nothing in this section is weakened by its absence: the statement is simply conditional.*

3.5. Provenance.

Every bound above is either an instance of a published theorem, an instance of one that must first be restated for the variant used here, or without a counterpart in the literature. Table 2 records which.

Bound	Status	What is needed
Lemma 3.3	new, but elementary	Divisibility by Z_Ω is a linear condition; power aggregation meets a proper subspace in at most $\lvert\mathcal{C}\rvert - 1$ points (Remark 1.2). Independent coefficients would give $1/\lvert\mathbb{K}\rvert$.
Lemma 3.4	standard	Schwartz-Zippel on a polynomial of degree at most δN.
Lemma 3.5	restatement required	Proximity gaps [Ben+20] for the curve traced by the nested powers of (μ_1, μ_2), of degree $\lvert\Gamma\rvert \cdot \max_i n_i$, rather than for an affine space. The published statement covers low-degree parametrised curves; the degree here is what enters $\varepsilon_{\mathrm{batch}}$.
Lemma 3.6	restatement required	FRI folding [Ben+18, Ben+20] with a per-layer arity $2^{m_{i-1}-m_i}$ rather than a fixed one, and with early stopping at layer L.

Bound	Status	What is needed
Lemma 3.7	standard	Each query lands on a disagreement point with probability at least θ.
Proposition 3.9	assembly, plus one restatement	BCS [BCS16] applied to capped arity-a Merkle trees (Definition 1.10); the passage from round-by-round to state-restoration soundness; and the accounting of grinding. The constant $c(Q, r)$ is to be pinned.
$\theta < 1 - \sqrt{\rho}$	assumption	The bounds hold in the provable Johnson regime. The configuration's proximity parameter may be set beyond it, in the conjectured regime up to $1 - \rho$, which is a separate assumption and must be declared as one.

TABLE 2. Provenance of the bounds of Section 3.3.

Two entries deserve emphasis. Lemma 3.5 is the one place where ZisK's choices change the shape of a published statement rather than its parameters, and it is the bound we are least able to quote. And the last row is not a bound at all but a modelling decision: whether the analysis is carried out in the regime where FRI is proved sound, or in the regime where it is conjectured to be, is the largest single assumption in this document, and it is the one a reader comparing security levels across zkVMs should look at first.

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